

PingPal - Requirements Document

Functional Requirements

FR1. The system shall allow a player to register with email, password, display name, city and skill level and it should allow to create the account.

FR2. The system shall allow a registered player to log in and see their profile (name, city, skill level).

FR3. The system shall display all registered players on the Find Players page with name, city and skill level.

FR4. The system shall allow a player to filter the player list by city using a search box.

FR5. The system shall display all coaches with name, city, bio, experience, speciality and hourly rate.

FR6. The system shall allow a player to filter coaches by city.

FR6a. The system shall list only coach profiles that have been added by the PingPal team. Coaches do not register or log in themselves. The PingPal team checks the coach's identity and experience, then adds the profile (name, city, bio, experience, speciality, hourly rate) to the database. This is what "verified" means in S5.

FR7. The system shall allow a player to submit a booking request to a coach with their name, email, preferred date and message.

FR8. The system shall allow a player to create a session with title, venue, city, date and max players.

FR9. The system shall display all upcoming sessions on the Sessions page.

FR10. The system shall allow two players to exchange messages using their display names.

FR11. The system shall validate all form inputs and show a clear error message when required fields are missing.

FR12. The system shall store all data persistently in a PostgreSQL database.

Non-Functional Requirements

- **Performance:** First page usable within 2 seconds on a 4G connection. API responses under 500ms for standard queries.

- **Accessibility:** Keyboard navigation works on all pages. All form inputs have labels. Colour contrast meets WCAG AA standard.
 - **Security:** Passwords stored as BCrypt hashes. No plain-text passwords in database or repository. API endpoints return only public data.
 - **Privacy:** No sensitive personal data exposed in API responses. Email addresses not shown on public player cards.
 - **Availability:** App is live at pingpal-production.up.railway.app during class hours. A failed deploy triggers a rollback the same day.
 - **Usability:** A new user can register and find a player within 3 minutes without instructions.
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User Stories

S1 · Player Registration

As a table tennis player in Estonia, I want to create an account with my skill level and city, so that other players can find and match with me.

Acceptance criteria:

- Given the register page is open, when I fill in all fields and click Register, then I see "Registered: [email]" and am redirected to Find Players
 - Given the register page is open, when I submit with an already used email, then I see "Email already exists"
 - Given the register page is open, when I leave any field empty and click Register, then I see a red error and nothing is submitted
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S2 · Player Login

As a registered player, I want to log in to my account, so that I can access my profile and the platform features.

Acceptance criteria:

- Given the login page is open, when I enter correct email and password and click Login, then I see "Welcome, [name]!" with my city and skill level
 - Given the login page is open, when I enter a wrong password, then I see "Invalid email or password"
 - Given the login page is open, when I enter an email that does not exist, then I see "Invalid email or password"
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S3 · View All Players

As a player, I want to see all registered players when I open the Find Players page, so that I can browse potential opponents.

Acceptance criteria:

- Given the Find Players page loads, when no search is applied, then I see all players with name, city and skill level
 - Given no players are registered yet, when the page loads, then I see "No players found."
 - Given players exist, when the page loads, then each card shows name, city and skill level
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S4 · Search Players by City

As a player, I want to filter players by city, so that I can find opponents who are near me.

Acceptance criteria:

- Given the Find Players page is open, when I type "Tallinn" and click Search, then I see only players from Tallinn
 - Given the Find Players page is open, when I search a city with no players, then I see "No players found."
 - Given I have filtered by city, when I click "Show All", then I see all players again
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S5 · Browse Coaches

As a player, I want to browse verified coaches with their rates and specialities, so that I can choose the right coach for my needs.

Acceptance criteria:

- Given the Coaches page loads, when no filter is applied, then I see all coaches with name, city, bio, experience, speciality and hourly rate
 - Given the Coaches page is open, when I search "Tallinn", then I see only Tallinn coaches
 - Given no coaches match the search, when I click Search, then I see "No coaches found."
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S6 · Book a Coach

As a player, I want to send a booking request to a coach, so that I can arrange a training session.

Acceptance criteria:

- Given the Coaches page, when I click "Book Session" on a coach, then I am taken to the booking page showing that coach's name and rate
 - Given the booking page, when I fill in all fields and click "Send Booking Request", then I see "Booking sent! [Coach name] will contact you soon." in green
 - Given the booking page, when I submit with any field empty, then I see a red error and booking is not saved
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S7 · Create a Session

As a player, I want to create a playing session at a venue, so that other players can see it and join me.

Acceptance criteria:

- Given the Sessions page, when I fill in all fields and click "Create Session", then I see "Session created: [title]" and the session appears in the list below
 - Given the Sessions page, when I submit without a date, then I see "Please fill in all fields!" and nothing is saved
 - Given sessions exist, when the page loads, then I see all sessions with title, venue, city, date and creator name
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S8 · Message Another Player

As a player, I want to send direct messages to another player, so that we can arrange a game without leaving the platform.

Acceptance criteria:

- Given the Messages page, when I enter my name and another player's name and click "Start Chat", then the chat window opens showing message history
 - Given the chat is open, when I type a message and click Send, then my message appears on the right side with my name
 - Given the chat is open, when the other player sends a message, then it appears on the left side and the chat refreshes automatically every 3 seconds
 - Given the Messages page, when I click "Start Chat" with an empty name field, then I see "Please enter both names"
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Events (from Event Storming)

Event (past tense)	Command that triggers it	External systems involved
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PlayerRegistered	SubmitRegistrationForm	PostgreSQL (save user)
PlayerLoggedIn	SubmitLoginForm	PostgreSQL (verify password hash)
PlayerListLoaded	OpenFindPlayersPage	PostgreSQL (fetch all players)
PlayerSearched	ClickSearchButton	PostgreSQL (filter by city)
CoachListLoaded	OpenCoachesPage	PostgreSQL (fetch all coaches)
CoachSearched	ClickSearchButton	PostgreSQL (filter coaches by city)
BookingRequested	ClickSendBookingRequest	PostgreSQL (save booking)
SessionCreated	ClickCreateSession	PostgreSQL (save session)
SessionListLoaded	OpenSessionsPage	PostgreSQL (fetch all sessions)
MessageSent	ClickSendInChat	PostgreSQL (save message)
MessageHistoryLoaded	ClickStartChat	PostgreSQL (fetch messages between two players)

MILESTONES

Milestone 1 — Foundation

Goal: Working Spring Boot app connected to database

- Spring Boot project created with Maven

- PostgreSQL database running in Docker
 - User registration and login working with BCrypt password hashing
 - Basic Spring Security configured
 - Project pushed to GitHub with CI/CD pipeline
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Milestone 2 — Core Player Features

Goal: Players can find each other

- Player profiles with city, skill level and display name
 - Find Players page with city search and filter
 - All players visible with cards showing name, city and skill level
 - Show All and Search buttons working
 - Data persisting in PostgreSQL
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Milestone 3 — Coach Marketplace

Goal: Players can discover and book coaches

- Coach profiles with bio, hourly rate, experience and speciality
 - Coaches page with city search filter
 - Booking form with name, email, date and message
 - Booking saved to database with PENDING status
 - Success and error messages shown to user
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Milestone 4 — Sessions and Messaging

Goal: Players can organise games and communicate

- Sessions page with create form and upcoming sessions list
 - Form validation — error shown if any field is empty
 - Sessions saved to database and displayed in real time
 - Messages page with player-to-player chat
 - Chat auto-refreshes every 3 seconds
 - Message history persisted in database
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Milestone 5 — Production Deployment

Goal: App is live and accessible to real users

- Dockerfile created for containerised deployment
- App deployed to Railway cloud platform
- PostgreSQL managed database on Railway
- GitHub CI/CD — every push auto-deploys
- Live URL working: pingpal-production.up.railway.app
- Beta test with real Estonian table tennis players

